

DIGITAL ELECTRONICS

Data Sheets

Activity 1.1.7 — deliverables and answers

Four Steps

01

Print five datasheets

Pages 1–2 only: connection diagram and function table

02

Identify part numbers

Five unfamiliar parts, their function and manufacturer

03

Match gate symbols

Shape and bubble to the matching 74LS device

04

Name package styles

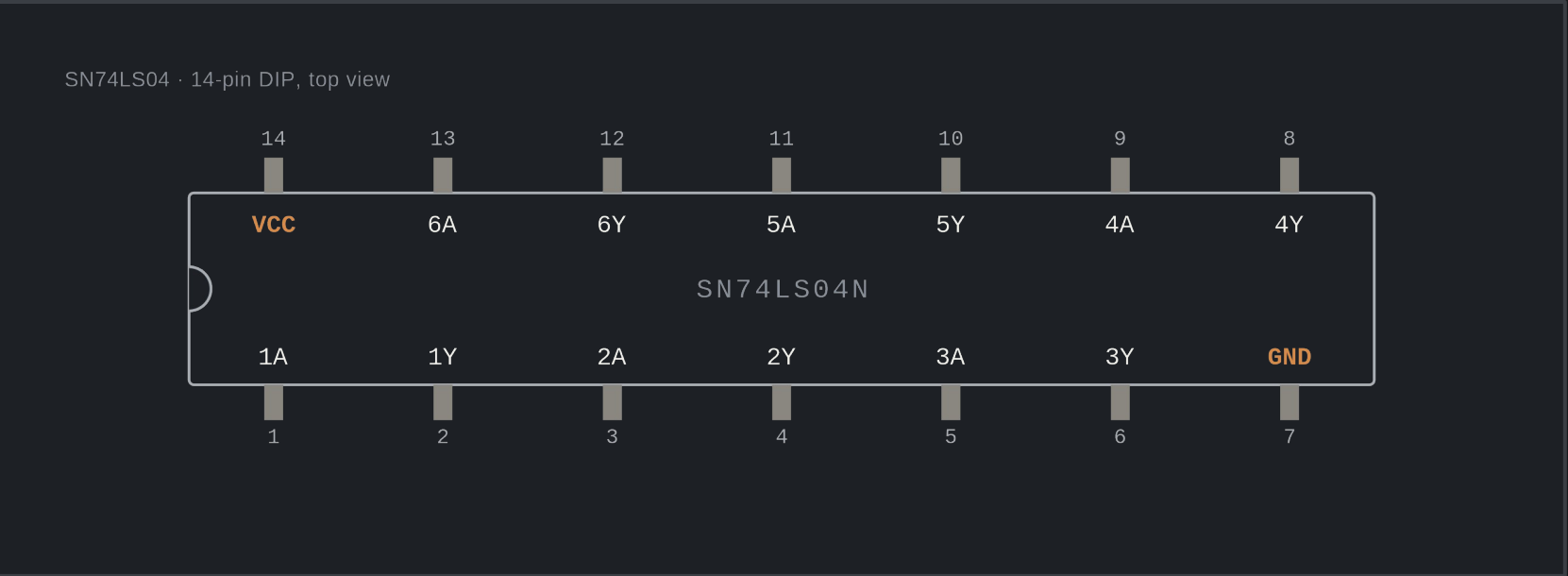
Five abbreviations expanded, with a sample image each

STEP 1

Datasheets to Print

IC	FUNCTION	DATASHEET
74LS04	Hex Inverter Gates	ti.com · sn74ls04
74LS08	Quad 2-Input AND Gates	ti.com · sn74ls08
74LS32	Quad 2-Input OR Gates	ti.com · sn74ls32
74LS74	Dual positive-edge-triggered D flip-flops with preset and clear	ti.com · sn74ls74a
LM555	Timer	ti.com · lm555

What Pages 1–2 Give You



Page 1 — Connection diagram

Pin numbering, gate assignments, and where V_{CC} and GND land on the package.

SN74LS74A · function table (each flip-flop)

PRE	CLR	CLK	D	Q	$\overline{Q}$
L	H	X	X	H	L
H	L	X	X	L	H
L	L	X	X	H*	H*
H	H	↑	H	H	L
H	H	↑	L	L	H
H	H	L	X	Q_0	$\overline{Q}_0$

* Both outputs high while PRE and CLR are both low; not a stable state.

Page 2 — Function table

Input-to-output behavior, including clock, preset, and clear conditions for sequential parts.

STEP 2

Unfamiliar Part Numbers

PART	NAME / FUNCTION	MANUFACTURER
DM74LS00	Quad 2-input NAND gate	Fairchild Semiconductor
SN74LS02	Quad 2-input NOR gate	Texas Instruments
DM74LS75	4-bit bistable latch — transparent, enable-controlled	Fairchild Semiconductor
SN74LS86	Quad 2-input XOR gate	Texas Instruments
MAN6760	0.56" single-digit seven-segment LED display — common anode, red	Fairchild Semiconductor

NAMING CONVENTION

SN

Texas Instruments

DM

Fairchild / National Semiconductor

Same die, different house prefix — the digits after it (74LS00, 74LS86) are what identify the logic function.

STEP 3

Gate Symbol to 74LS Part

SYMBOL	GATE	PART
Flat-back (D) shape · 3 inputs · no bubble	3-input AND	74LS11
Flat-back (D) shape · 3 inputs · bubble	3-input NAND	74LS10
Flat-back (D) shape · 4 inputs · no bubble	4-input AND	74LS21
Flat-back (D) shape · 4 inputs · bubble	4-input NAND	74LS20
Curved-back (shield) shape · 3 inputs · bubble	3-input NOR	74LS27

Reading a Gate Symbol

Shape

Flat back means AND family

Curved shield back means OR family.

Inputs

Count the input lines

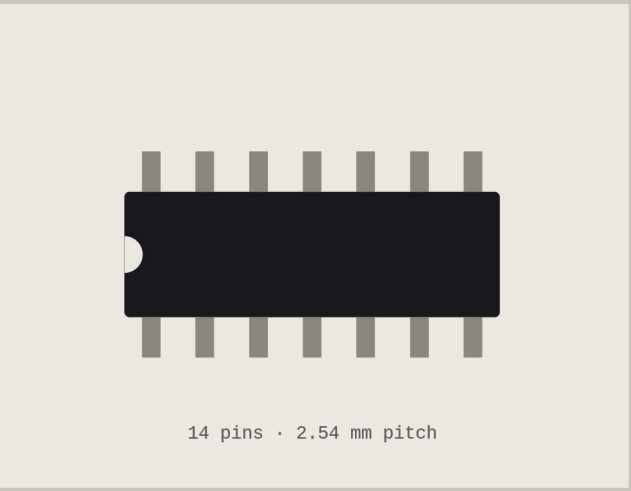
Input count separates 74LS11 from 74LS21, 74LS10 from 74LS20.

Bubble

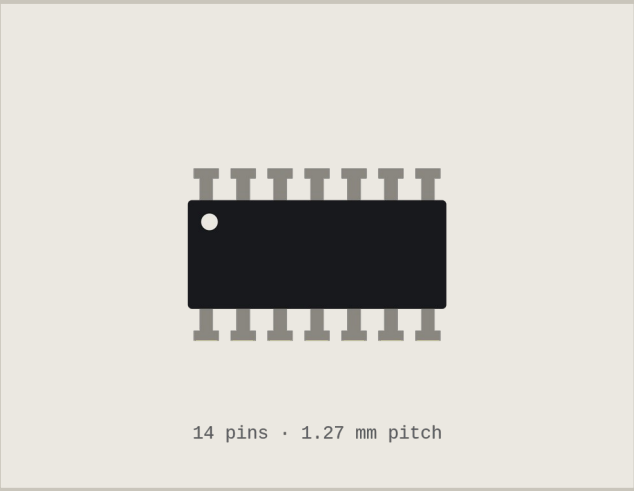
A bubble inverts the output

AND becomes NAND; OR becomes NOR.

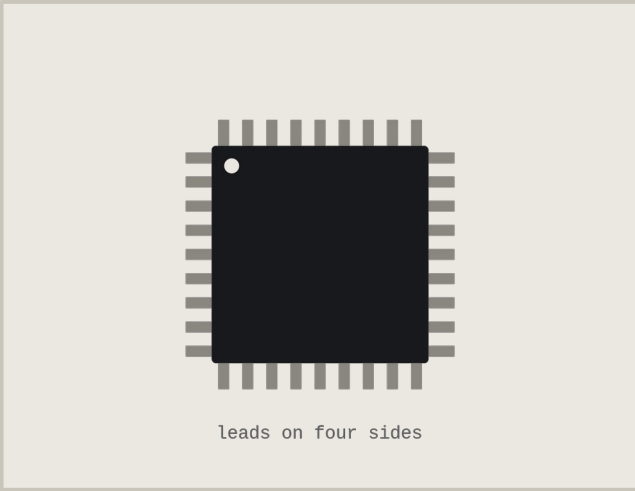
IC Package Styles



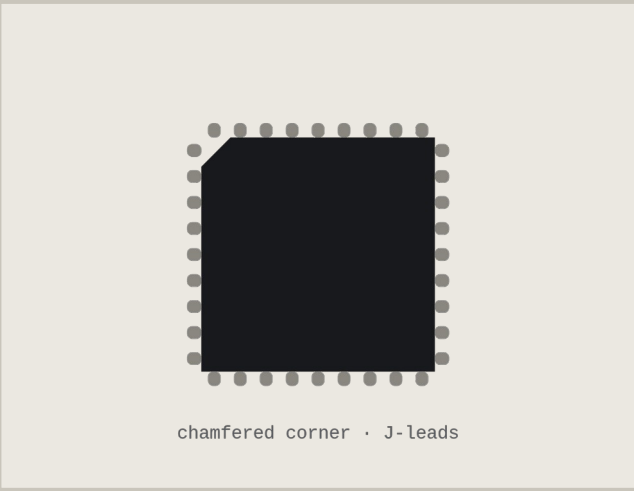
DIP
Dual In-line Package



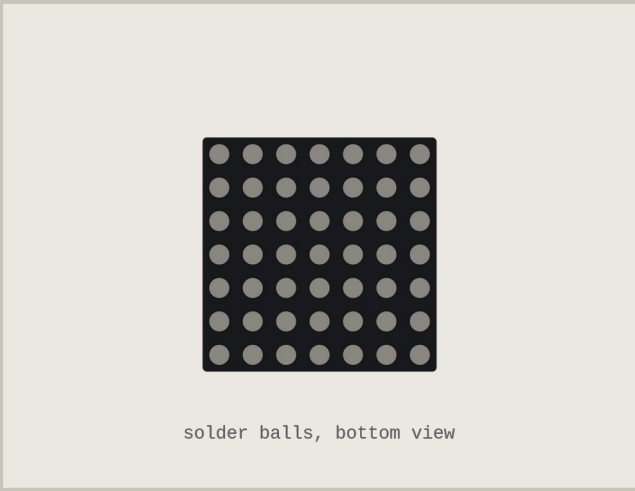
SOIC
Small Outline
Integrated Circuit



QFP
Quad Flat Package



PLCC
Plastic Leaded Chip
Carrier



BGA
Ball Grid Array

What Gets Turned In

- 01 Ten printed pages — pages 1–2 for each of the five ICs
- 02 Part-number table with function and manufacturer
- 03 Gate-symbol table with matching 74LS part numbers
- 04 Package-style table with a sample image for each